CONSEQUENCES OF THE EXISTENCE OF A LEFT ADJOINT FOR A COCONTINUOUS FUNCTOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The functors on presheaves attached to an adjoint pair. Same chapter, Lemma lemma-adjoint-functors and Lemma lemma-adjoints-u. Identify the left Kan extension with the direct image along the left adjoint and show that it is a right adjoint.
- Limits of sheaves and exactness of sheafification. Same chapter, Lemma lemma-limit-sheaf and Lemma lemma-sheafification-exact. Transfer the commutation with finite limits from presheaves to sheaves.
- The pushforward along a continuous functor and its adjoint. Same chapter, Lemma lemma-pushforward-sheaf and Lemma lemma-adjoint-sheaves. Supply the left adjoint in the continuous case.

Source reference. Original lemma 00XO: sources/stacks-sites/pinned-original-source.tex, source lines 5205–5205. Navigation pointer invoked at proof line 5205; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00XR: sources/stacks-sites/pinned-original-source.tex, source lines 5206–5206. Navigation pointer invoked at proof line 5206; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 09VQ: sources/stacks-sites/pinned-original-source.tex, source lines 5225–5225. Navigation pointer invoked at proof line 5225; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0038: sources/stacks-sites/pinned-original-source.tex, source lines 5229–5229. Navigation pointer invoked at proof line 5229; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00W2: sources/stacks-sites/pinned-original-source.tex, source lines 5232–5232. Navigation pointer invoked at proof line 5232; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00VE: sources/stacks-sites/pinned-original-source.tex, source lines 5233–5233. Navigation pointer invoked at proof line 5233; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00WJ: sources/stacks-sites/pinned-original-source.tex, source lines 5236–5236. Navigation pointer invoked at proof line 5236; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00WW: sources/stacks-sites/pinned-original-source.tex, source lines 5242–5242. Navigation pointer invoked at proof line 5242; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00WX: sources/stacks-sites/pinned-original-source.tex, source lines 5242–5242. Navigation pointer invoked at proof line 5242; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00XX: sources/stacks-sites/pinned-original-source.tex, source lines 5143–5143. Navigation pointer invoked at proof line 5143; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ and $\mathcal{D}$ be sites. Let $g : Sh(\mathcal{C}) \to Sh(\mathcal{D})$ be the morphism of topoi associated to a continuous and cocontinuous functor $u : \mathcal{C} \to \mathcal{D}$, see Lemmas 00XO and 00XR.*

1. *If $w : \mathcal{D} \to \mathcal{C}$ is a left adjoint to u, then*
   (a) *$g_!\mathcal{F}$ is the sheaf associated to the presheaf $w^p\mathcal{F}$, and*
   (b) *$g_!$ is exact.*
2. *if w is a continuous left adjoint, then $g_!$ has a left adjoint.*
3. *If w is a cocontinuous left adjoint, then $g_! = h^{-1}$ and $g^{-1} = h_*$ where $h : Sh(\mathcal{D}) \to Sh(\mathcal{C})$ is the morphism of topoi associated to w.*

Proof. Recall that $g_!\mathcal{F}$ is the sheafification of $u_p\mathcal{F}$. Hence (1)(a) follows from the fact that $u_p = w^p$ by Lemma 09VQ.

To see (1)(b) note that $g_!$ commutes with all colimits as $g_!$ is a left adjoint (Categories, Lemma 0038). Let $i \mapsto \mathcal{F}_i$ be a finite diagram in $Sh(\mathcal{C})$. Then $\lim \mathcal{F}_i$ is computed in the category of presheaves (Lemma 00W2). Since w^p is a right adjoint (Lemma 00VE) we see that $w^p \lim \mathcal{F}_i = \lim w^p\mathcal{F}_i$. Since sheafification is exact (Lemma 00WJ) we conclude by (1)(a).

Assume w is continuous. Then $g_! = (w^p\ )^\# = w^s$ but sheafification isn't necessary and one has the left adjoint w_s, see Lemmas 00WW and 00WX.

Assume w is cocontinuous. The equality $g_! = h^{-1}$ follows from (1)(a) and the definitions. The equality $g^{-1} = h_*$ follows from the equality $g_! = h^{-1}$ and uniqueness of adjoint functor. Alternatively one can deduce it from Lemma 00XX. □